

Product Returns under Perfect Competition

Draft notes

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This note records a competitive private-ownership economy with a continuum of buyers and a continuum of identical price-taking firms. The first section is the environment without returns. The second allows inspection and a refund. Unlike the incidence notes, the refund is not restricted to equal the list price.

1 No returns

1.1 Setting

The economy has a continuum of consumers, each with a numeraire endowment ω_i , and a continuum of identical price-taking firms. Consumer i 's income is ω_i plus her share of industry profit.

Each firm produces at constant unit cost c . The good cannot be returned.

Consumers have unit demand, quasi-linear preferences, and an outside option of zero. They differ in their value v_i for the good, known when they buy.

Consumers take a price p as given. Each buys at most one unit, or not at all. Buying yields $v_i - p$. She buys when $v_i \geq p$.

A firm that sells y units earns $(p - c)y$. A price-taker then compares p with c .

Industry output equals the measure of purchases.

1.2 Equilibrium

A competitive equilibrium is a price p , purchases, and industry output y such that consumers optimize given p and their income, firms maximize profit given p , and the market clears. Both sides see the same p .

Consumer i buys if and only if $v_i \geq p$. A firm that takes p as given earns $(p - c)y$, so it supplies nothing if $p < c$, any scale if $p = c$, and infinite output if $p > c$. Market clearing requires y to equal the measure of buyers. Infinite supply cannot clear against a finite mass of consumers, and zero supply cannot clear if a positive measure wants to buy. Whenever demand is positive, therefore $p = c$ and industry profit is zero. Firms then meet that demand.

2 Returns

2.1 Setting

The good can now be purchased, inspected, and returned. Checkout price and refund are two prices. Each firm still produces at cost c and now incurs a unit handling cost h on every returned item. Returned units are salvaged rather than restocked.

Consumers still have unit demand, quasi-linear risk-neutral preferences, and an outside option of zero. They differ in return hassle κ_i , the disutility of sending the good back, and in the distribution G_i of match value v realized after purchase. Consumer i knows (κ_i, G_i) when she buys, but not v .

Consumers take a checkout price p and a refund R as given. Each buys at most one unit, or not at all, then draws $v \sim G_i$ and decides whether to keep the item or return it.

If she has bought at (p, R) , keeping yields $v - p$ and returning yields $R - p - \kappa_i$. She therefore keeps if and only if $v \geq R - \kappa_i$. The keep rule depends on the refund and on hassle, not on the checkout price. Expected surplus from buying is

$$cs_i(p, R) = \mathbb{E}_{v \sim G_i} [\max(v - p, R - p - \kappa_i)] = -p + \mathbb{E}_{v \sim G_i} [\max(v, R - \kappa_i)].$$

She buys when that value is nonnegative. The reservation checkout price for a given refund, $p_i(R) = \mathbb{E}[\max(v, R - \kappa_i)]$, is the largest p at which she is willing to buy. Heterogeneity therefore cannot be collapsed into a single willingness to pay that is independent of the refund. Consumer i 's return probability at refund R is $\phi_i(R) = G_i(R - \kappa_i)$.

A firm that sells y units collects p on each sale, refunds R and incurs h on the share r that come back, and therefore earns

$$\pi = (\mu - c)y, \quad \mu = p - r(R + h),$$

where μ is the effective margin. A price-taker then compares μ with c .

Industry output equals the measure of purchases. The returned share r is the mean of $\phi_i(R)$ among those who buy, and who buys depends on (p, R) .

2.2 Equilibrium

A competitive equilibrium is a pair (p, R) , purchases, a return rate r , and industry output y such that consumers optimize given (p, R) and their income, firms maximize profit given (p, R) and r , and the market clears. Both sides see the same (p, R) .

Consumer i buys if and only if $cs_i(p, R) \geq 0$, and returns with probability $\phi_i(R)$. The return rate $r(p, R)$ is the mean of $\phi_i(R)$ among those buyers. A firm takes (p, R) and r as given and earns $(\mu - c)y$, so it supplies nothing if $\mu < c$, any scale if $\mu = c$, and infinite output if $\mu > c$.

Market clearing requires y to equal the measure of buyers, and the r in the effective margin must be $r(p, R)$. Infinite supply cannot clear against a finite mass of consumers, and zero supply cannot clear if a positive measure wants to buy. Whenever demand is positive, therefore

$$p - r(p, R)(R + h) = c. \tag{1}$$

As long as this holds, the pair is a competitive equilibrium. Consumers take (p, R) as given and follow the buy and keep rules, which determines demand and $r(p, R)$. Then $\mu = c$, so every firm is willing to sell any scale, and industry can meet that demand. Buyers optimize, sellers optimize, and the market clears.

(1) is one equation in two prices, so many pairs are competitive equilibria. Consumers keep if and only if $v \geq R - \kappa_i$, while a return costs h . Efficiency therefore requires $R = -h$ and $p = c$. That pair satisfies (1), so the efficient allocation can be sustained as a competitive equilibrium. At other pairs satisfying (1) consumers return too often or too rarely.